

Test Report: Radial Wooden Plain Bearing

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Project: Substrate Mixer and Portioning Device

Component: Plain Bearings of Agitator

Prototype Version: Final Application

PART I — TEST PLAN

1. Research Hypothesis

A low-cost wood-made radial journal bearing is sufficient to support a rotating steel drive shaft in a manual agitator. It is hypothesized that wood provides adequate structural stability and sufficiently low friction ($\mu < 0.3$) for low-frequency operation ($f < 1$ Hz), eliminating the need for expensive rolling-element bearings.

2. Test Objective

Determine the experimental friction coefficient between the steel shaft and the wooden bearing and compare these results with the values established in literature.

3. Measurement Methodology

- **Instruments:** Analog force gauge (spring scale).
- **Procedure:** Measurement of tangential force F_{meas} at the crank lever ($l = 0.09$ m) during steady rotation.
- **Experimental Calculation of Friction Coefficient μ :**

$$\mu = \frac{F_{meas} \cdot l_{lever}}{F_N \cdot r_{shaft}}$$

(Bender & Göhlich, 2020, p. 251)

4. Success Criteria

The bearing is considered suitable if $\mu < 0.3$ under operational conditions (Wittel et al., 2013, p. 535).

PART II — TEST EXECUTION DATA

Radial Load Calculation (F_N)

The radial load is determined by the total mass of the agitator assembly:

- Component 1 3.0 kg; Component 2 2.5 kg; Total Mass (m) 5.5 kg

$$F_N = m \cdot g = 5.5 \text{ kg} \cdot 10 \text{ m/s}^2 \approx \mathbf{55 \text{ N}} \quad (1)$$

Table 1: Fixed physical and geometric parameters of the agitator assembly.

Parameter	Symbol	Value	Unit
Radial Load (Static)	F_N	55	N
Crank Lever Length	l_{lever}	0.09	m
Shaft Radius	r_{shaft}	0.0135	m

Table 2: Comparative results of friction coefficient μ before and after bore modification.

No.	Condition	F_{meas}	M	μ	Notes
1	Initial	5.0 N	0.45 Nm	0.61	Baseline; high parasitic friction.
2	Post-Boring	3.0 N	0.27 Nm	0.36	Dry run; clearance optimized.

PART III — TEST REVIEW

1. Observations

- **Test 1 (Initial):** Significant acoustic emissions (squeaking). High resistance due to shaft interference with the drum housing.
- **Test 2 (Modified Bore):**
 - *Acoustics:* Squeaking significantly reduced, though not eliminated.
 - *Deceleration:* The "coast-down" behavior improved drastically; after a single impulse, the shaft completes 6–7 full revolutions before stopping.
 - *Mechanical Imbalance:* Due to processing inaccuracies (paddles not perpendicular), the shaft exhibits significant imbalance. This causes the required input force to fluctuate between 0 N and 15 N depending on the angular position of the crank. The recorded 3.0 N represents an average force.

2. Data Interpretation

Test 1: Results are invalid for bearing assessment due to dominant parasitic friction at the housing interface.

Test 2: A friction coefficient of $\mu \approx 0.36$ was achieved. Precise measurement is hindered by shaft imbalance, as the varying center of gravity overlaps with the friction torque. However, the excellent coast-down performance (6–7 revolutions) qualitatively confirms that the actual friction in the wooden bearing is very low and likely satisfies the target requirements.

3. Knowledge Gained

- **Tolerance:** Oversize drilling is essential to isolate bearing friction. Grease lubrication also results in less friction and noticeably reduces noise emissions.
- **Manufacturing:** Agitator imbalance significantly affects torque consistency, but does not compromise the functional viability of the wooden bearing.
- **Performance:** A dry friction value of approximately $\mu \approx 0.3$ is achievable, which is sufficient for low-tech and low-cost requirements.

4. Hypothesis Evaluation

[X] Confirmed

Explanation: Structural stability is verified. Friction requirements are met once parasitic drag is eliminated. The prototype demonstrates that plain wooden bearings are a viable replacement for industrial bearings in this specific low-frequency application.

5. Next Development Steps

- **Conclusion:** The development of low-cost plain bearing is complete. No further structural modifications are required.

References

- Bender, B., & Göhlich, D. (2020). *Dubbel Taschenbuch für den Maschinenbau 1: Grundlagen und Tabellen* (26th ed.). Springer. <https://doi.org/10.1007/978-3-662-59711-8>
- Wittel, H., Muhs, D., Jannasch, D., & Vossiek, J. (2013). *Roloff/Matek Maschinenelemente* (21st ed.). Springer. <https://doi.org/10.1007/978-3-658-02327-0>